

3.	a) What is instruction? What is the main difference between an opcode and an operand in a microprocessor? Explain with examples.	CO3	3
	b) Consider the following scenario: 1. Tanveer and Shihab are good friends. Tanveer was preparing himself for the midterm exam, sitting in the library. At that time, Shihab called Tanveer and asked him to come and play at Central Field. Tanveer avoided him and said he would come after 30 minutes after completing his	CO2	4

assignment. At the same time, Tanveer's teacher called and asked to meet to arrange a special class. Tanveer went to meet without avoiding. Shihab noticed something in the pin diagram of the microprocessor (intel 8086): the timing of the interrupt was also the same. Some interrupts are avoided by the microprocessor, while some are not.

Now write and discuss the pins responsible for the interrupts.

2. DMA stands for Direct Memory Access. It is designed by Intel to transfer data at the fastest rate. It allows the device to transfer the data directly to/from memory without any interference of the CPU.

Now, discuss the pins responsible for DMA techniques in 8086 μ p.

OR,

In an arithmetic operation involving the 8086 Microprocessor, describe how the general-purpose registers AX, BX, CX, and DX are utilized. Discuss the specific roles each register plays during the execution of the operation. Provide an example arithmetic operation and illustrate the step-by-step involvement of these registers in the process.

c) Write an assembly language program to compute $Y = (5+3) * (2+1)$

CO2

3

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE, Midterm Examination, Spring 2024

Course Code: CSE-3529

Course Title: System Analysis and Design

Total marks: 30, Time: 1 hour 30 minutes.

[Answer all the questions. Figures in the right-hand margin indicate full marks]

1. a) Define system. Discuss the stages in building system. Who is system analyst? 5 CO1
Explain role, responsibilities and qualities of a system analyst.
- b) TechGadget Zone, a prominent electronics retailer, has announced a special promotion named "TechFiesta Mega Sale," offering a 30% discount on all electronic gadgets for a limited time, starting from 1st January to 15th January 2024. "How does the TechFiesta Mega Sale promotion at TechGadget Zone exemplify the integration of strategic, tactical, and operational information? Provide specific details on how the strategic decision-making, the tactical planning and the operational considerations addressed to ensure a smooth and successful implementation during the specified period." 5 CO2
2. a) Discuss Rapid Application Development (RAD) and Joint Application Development (JAD) methodologies with their advantages and disadvantages. 5 CO1
OR (of 2a)
What is SDLC? Briefly describe the different phases of SDLC.
- b) A software development company currently offered to develop following 5 CO2
projects:
 - i. A new accounting software for a client. The client has provided detailed requirements upfront and expects the software to be delivered in a short period of time.
 - ii. Another project is a new social networking platform. The company wants to adapt to changing market needs quickly and release new features iteratively to gather user feedback.

Above the scenario which SDLC model would adopt for developing both projects? Justify your answer.
3. a) Define project management. Describe the execution phase of project 5 CO1
management.
- b) A very basic cash flow projection for an IT Projects is, in the Year 0,1,2,3 costs 5 CO2
are 50000, 8000, 10000, 14000 and benefits are 0, 25000, 30000, 48000. Find out the followings for the above conditions:
 - (i) Return on Investment
 - (ii) Break-Even Point

OR (of 3b)

Describe cost-benefit analysis. Briefly describe the different metrics to measure projects feasibility in terms of economic value.

INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

Department of Computer Science & Engineering (CSE)

BSc in CSE, Mid-Term Examination, Spring 2024

Course Code: CSE-3527 Course Title: Compiler

Full Marks: 30 Time: 1 Hour 30 minutes

[Answer all of the following questions. Some questions may have an option.
(Figures at right margin illustrate the marks (M), course objectives (COs), and bloom's taxonomy (DL))]

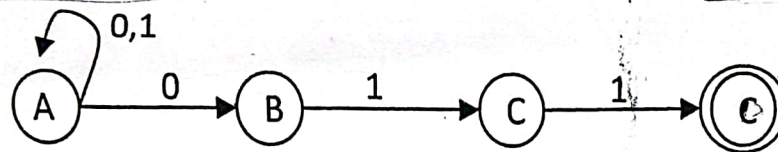
No.	Questions	M	COs	DL
1. a)	Elucidate the block diagram to define a tool to convert an input source program into an output target program and contrast it with the interpreter. What is the significant role of this tool in a language processing system, and illustrate it with an appropriate figure of a language processing system?	4 (2+2)	CO1	U
b)	Translate the following C programming assignment statement step by step in compiler phases. Draw the diagram of the compiler's phases with output annotation of each phase and symbols table. <code>net_salary = basic + house_rent - basic * tax_rate;</code> [where the value of <code>tax_rate</code> is fractional]	6 (4+2)	CO1	An
2. a)	Illustrate the concept of ambiguity and how it impacts on a compiler design? Consider the following CFG G_1 : $\begin{aligned} \text{Exp} &\rightarrow \text{Exp} + \text{Exp} \mid \text{Exp} - \text{Exp} \\ \text{Exp} &\rightarrow \text{Exp} * \text{Exp} \mid \text{Exp} / \text{Exp} \\ \text{Exp} &\rightarrow 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9 \\ \text{Exp} &\rightarrow \text{id} \end{aligned}$ Justify whether the above grammar is ambiguous using its input string "6-5+2*1/id". If it is ambiguous, convert it into the unambiguous grammar and construct the syntax tree for the same input string.	5 (2+3)	CO2	Ap
	Or What do you mean by synthesized attributes? Consider the following CFG G_2 : $\begin{aligned} E &\rightarrow E + T \mid E - T \\ E &\rightarrow T \\ T &\rightarrow 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9 \end{aligned}$ i) Write the syntax directed definition using semantic rules for the above CFG G_2 and Create an annotated parse tree for the input string "9-5+2-1" ii) Generate leftmost and rightmost derivation for the input string: "5+1-4+8"	5 (1+2+2)	CO2	An
b)	Illustrate the roles of lexical analyzer in a compiler with schematic diagram. What is the main advantage of separating the analysis phase of compiling into lexical analysis and parsing? Identify the <u>tokens</u> , <u>lexemes</u> and <u>patterns</u> of the following C programming function in a table:	5 (2+1+2)	CO2	U

```
long factorial (int n) {
    if (n==1)
        return 1;
    else
        return n*factorial(n-1);
}
```


3. a) Why finite automata is important in a compiler construction? Write the regular expression to define the identifier of C programming language and construct the ϵ - NFA and DFA using that regular expression. 5 CO2 An (1+1+2+1)

Or

Contrast DFA with NFA. Construct the equivalent NFA for the regular expression $RE_1: (a|b)^*abb(a|b)$ directly and Convert the following NFA to DFA. 5 CO2 An (1+2+2)



- b) Define syntax directed translation of *if-else* conditional statement of C programming language using appropriate grammar. What is the main factor to identify the types of grammar? Identify the regular grammars, right-linear grammars, left-linear grammars, context free grammars, context sensitive grammars and useless non-terminals in the following grammars. 5 CO2 Ap (2+3)
- [Some grammar may fit more than one category].

G3:

$S \rightarrow aM$

$M \rightarrow Pb$

$Ma \rightarrow bPc$

$Pc \rightarrow ac$

G4:

$S \rightarrow aX | \epsilon$

$X \rightarrow aY$

$Y \rightarrow b$

G5:

$A \rightarrow aA | \epsilon$

$B \rightarrow Ab | b$

$C \rightarrow c$

G6:

$S \rightarrow Aa | AbTy_1$

$B \rightarrow Cb | Dd$

$C \rightarrow c$

END

CFG
CSG

$A \rightarrow a^*$

$B \rightarrow Ba$
 $B \rightarrow \alpha a$

International Islamic University Chittagong
Department of Computer Science & Engineering
Program: B. Sc (Engg.) in CSE Semester: 5th
Mid-Term Examination, Spring 2024

Course Code: CSE-3521

Course Title: Computer Architecture

Time: 1 hour and 30 minutes

Total Marks: 30

[Parts of the same question must be answered following the serial]

- Q.1.** a. Define the terms CPI and CPU clock cycle. "MIPS can vary inversely with performance"- verify this statement with appropriate example.
 b. Suppose we have two implementations of the same instruction set architecture. Machine A has a clock cycle time of 1 ns and a CPI of 3.0 for some program, and machine B has a clock cycle time of 2 ns and a CPI of 2.2 for the same program. Which machine is faster for this program, and by how much?

Or

Codes	Instruction Counts for instruction class(in billions)		
	A	B	C
Compiler 1	4	2	3
Compiler 2	7	6	5

Instruction Class	CPI for this Class
A	1
B	2
C	3

Assume that the computer's clock rate is 5 GHz. Which code sequence will execute faster according to MIPS? According to execution time?

- c. Subtract (35-7) using 2's complement.
- Q.2.** a. Define instructions. What is the major difference between the variables of a programming language and registers?
 b. "Good Design Demands Good Compromises"-is one of the design principles of MIPS. Give proper explanations with example.
 c. Translate the following MIPS Assembly Language instructions into Machine Language. You must specify the MIPS fields and Instruction Format used.
 (i) add \$s1, \$t5, \$t6.
 (ii) bne \$s2, \$s4, 100
 (iii) sw \$s1, 100(\$s2)
 d. What are the reasons for considering 32 bit registers each of size 32? Explain

Or

What is spilling register? When it is necessary to spill register?

- Q.3** a. Briefly describe different addressing modes in MIPS.
 b. Show the IEEE 754 binary representation of the number -0.75_{10} in single and double precision.

Or

Describe the first version of multiplication hardware using flowchart.

- c. Multiply $4_{10} \times 5_{10}$ using final version of multiplication algorithm.

Marks CLO DL
4 1 1

4 2 3
1x clock cycle time

CPI time = $I \times C$
clk cycle
7x3 = 21
6x2 = 12

01000101
00100010
01001001
20 12 13 3 0 0
0 0 1 0 1 0 0
1+1=2 1 1

3 1 2
3 2 3

So - to 6-15
16-23 to 8
24-31 to 0
17 to 10
18 to 11
19 to 12
20 to 13
21 to 14

2 1 1
4 2 3

4 2 3

11011
01011
101000
101011
11

International Islamic University Chittagong (IIUC)

Department of Computer Science and Engineering (CSE)

Mid-term Examination, Spring-2024

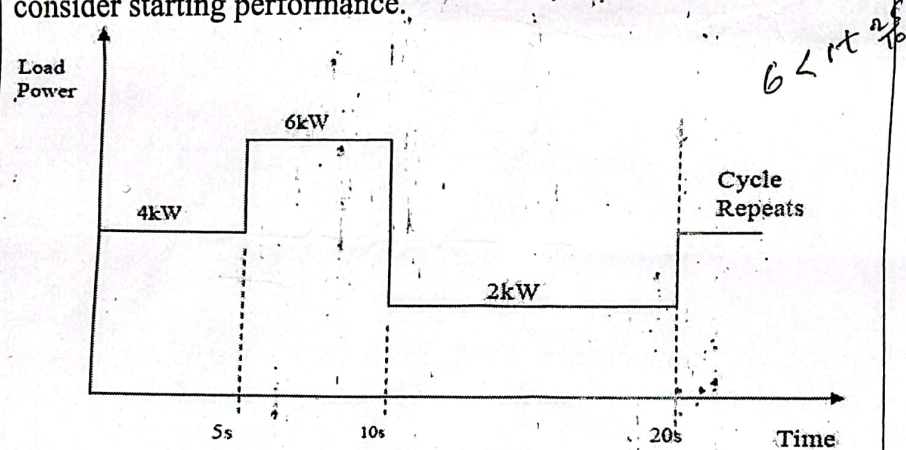
Program: BSc in CSE

Course Title: Electrical Drives and Instrumentation

Course Code: EEE-2421

Time: 1 hour 30 minutes

Full Marks: 30

1(a)	Define Machine. Describe the operation of motor with proper net sketch.	CO1	R	4
1(b)	The load torque profile shown in figure 1(b) was measured for a constant speed load. Choose an appropriately sized motor to drive this load assuming a service factor of 25%. You do not need to consider starting performance.  Figure: 1(b)	CO2	An	3
1(c)	What is motor duty class? Write the different types of motor duty class with appropriate timing diagram.	CO1	U	3

2(a)	Define transformer. Why transformer is rated on VA?	CO1	R	2
2(b)	A single phase transformer has 400 primary and 1000 secondary turns. The net cross sectional area of the core is 60 cm^2. If the primary winding be connected to a 50 Hz supply at 520V, calculate- i) the peak value of flux density in the core ii) the voltage induced in the secondary winding.	CO1	An	3
2(c)	A 230/460 V transformer has a primary resistance of 0.2Ω and reactance of 0.5Ω and the corresponding values for the secondary are 0.75Ω and 1.8Ω respectively. Find the secondary terminal voltage when supplying 10A at 0.8 power factor (p.f.) lagging.	CO2	An	5

OR

	A 50kVA, 4400/220-V, 50-Hz transformer has a high voltage winding resistance (R_1) of $3.45\ \Omega$ and a leakage resistance (X_1) of $5.2\ \Omega$. The low voltage winding resistance (R_2) is $0.009\ \Omega$ and the leakage resistance (X_2) is $0.015\ \Omega$. Find the equivalent winding resistance, reactance and impedance referred to the- i) high voltage side and ii) low voltage side.			
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3(a)	Draw the equivalent circuit of transformer with proper notation.	CO1	R	2
3(b)	Write the faraday's law of electromagnetic induction. A coil that has 700 turns develops an average induced voltage of 50V. What must be the change in the magnetic flux occur to produce such a voltage if the time interval for this change is 0.7s.	CO1	An	3
3(c)	A shunt generator delivers 450 A at 230V and the resistance of the shunt field and armature are $50\ \Omega$ and $0.03\ \Omega$ respectively. Calculate the generated emf. OR <i>In no-load test of single phase transformer, the following test data were obtained: Primary voltage: 220V; Secondary voltage: 110V; Primary Current: 0.5A; Power Input: 30W. Resistance of the primary winding = $0.6\ \Omega$</i> Find the following: i) The turn ratio ii) The magnetizing component of no load current iii) Its working (or loss) component iv) The iron loss.	CO2	An	5

International Islamic University Chittagong
Center for General Education
Mid-Term Examination, Spring 2024

Course Code: URED 3503 (URED 3101 for LLB)

Course Title: Political Thoughts & Social Behavior

Time: 1:30 Hours.

Full Marks: 30

[Answer all three of the following]
(All questions are of equal value)

- | | | |
|---|-----------|----------------------|
| 01. Explain the general guidelines that Islamic Shariah has given to run our political activities and then differentiate between the Islamic and conventional systems of politics | CLO
01 | Evaluate,
Explain |
| 02. Define <i>Shariah</i> and explain some characteristics of a constitution under Islamic Shariah. | CLO
02 | Explain |
| 03. What are the qualities of the chief executive of an Islamic State? Discuss them with some details. | CLO
02 | Remember |

Or

- | | | |
|--|-----------|----------|
| Evaluate the Legislative Organ of the Government under the <i>Shuratic</i> perspective and mention the functions of the legislative body of the state. | CLO
02 | Evaluate |
|--|-----------|----------|

International Islamic University Chittagong
Morality Development Program
Mid Term Examination, Spring -2024

5th Semester (other than Shari'ah faculty)

Course Code: MDP3505. Course Title: Concept of Moral Development II

Time: 1hr 30min

Marks: 30

Answer the following questions.

1. Describe the role of students in the formation of an ideal society in light of Islam. 10
2. a. What is the difference between dowry and Mahr in Islam? 3+3+4=10
b. Describe the negative effects of dowry from society.
c. Explain the role of students to eliminate dowry from society.
3. a. What is the difference between entrepreneurship and entrepreneur? 2+3+5=10
b. Write Characteristics of entrepreneurship.
~~c. Describe the reasons why multi-level marketing is prohibited in Islam.~~

Or

5x2=10

- a. Compare the differences between Islamic and Western concepts on entertainment.
- b. Identify misconceptions about Islam and write ways to remove the misconceptions.